Abdul Saboor

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PROFESSIONAL SUMMARY

Computer Science undergraduate at FAST NUCES specializing in Applied AI and LLM systems. Built and deployed multi-agent AI applications, RAG pipelines, and NLP models using LangChain, PyTorch, and FastAPI. Strong interest in production-grade ML systems, agent orchestration, and scalable AI applications, with a focus on building real-world, user-facing AI solutions.

TECHNICAL SKILLS

- **Languages:** Python, C/C++, JavaScript, TypeScript, SQL, x86 Assembly
- **AI / ML:** PyTorch, TensorFlow, Scikit-learn, Keras, NLP, RAG, LangChain, LangGraph
- **Data:** Pandas, NumPy, Matplotlib, PostgreSQL, Firebase
- **Frameworks & Tools:** FastAPI, Flask, Streamlit, Next.js, Node.js, Docker, Git, GitHub Actions
- **APIs & Cloud:** Google Gemini API, Groq API, Tavily API, Google Maps API, Vercel, Streamlit Cloud

EXPERIENCE

Full-Stack Developer Intern – A-Nets Cyber Solution *June 2026 – August 2026*

- Worked on full-stack web development, including responsive front-end interfaces, back-end functionality, RESTful APIs, and database integration.
- Handled debugging, testing, and technical documentation to support reliable, maintainable feature delivery.
- Contributed to development projects while collaborating with the team to deliver assigned features and solutions.

Lab Demonstrator – Programming Fundamentals *Aug 2025 – Dec 2025*

FAST – National University of Computer and Emerging Sciences, Karachi

- Taught programming fundamentals to 50+ students, improving average lab performance and code quality.
- Mentored students through debugging sessions, reducing common coding errors and improving problem-solving skills.
- Assisted in grading and feedback, ensuring consistent evaluation across assignments.

PROJECTS

StratosAI – Multi-Agent Startup Analyst

Python, LangGraph, LangChain, FastAPI, Next.js, Groq (LLaMA 3), Tavily API, ChromaDB

- Built a multi-agent AI system with 4 specialized agents (market research, competitor analysis, financial modelling, report synthesis) orchestrated via LangGraph StateGraph.
- Integrated Tavily API and Groq LLaMA 3 to generate structured market insights including TAM, competitor mapping, and risk analysis, with a FastAPI backend streaming progress via SSE and a Next.js visualization dashboard.

AI-Powered Adaptive Traffic Control System

Python, PyTorch, YOLOv8, Stable-Baselines3 (PPO), SUMO, FastAPI, React, WebSockets

- Built a real-time adaptive traffic control system using YOLOv8-based computer vision to estimate vehicle density and optimize signal timing dynamically.
- Trained a PPO reinforcement learning agent in SUMO and integrated a FastAPI + React dashboard with WebSockets for live traffic metrics and emergency vehicle signal preemption.

MediMind – LLM-Powered Health Assistant

Python, LangChain, Google Gemini API, RAG, FAISS, HuggingFace, Streamlit

- Built a RAG pipeline over medical PDFs using HuggingFace embeddings in FAISS, integrated with Google Gemini 2.5 Flash via LangChain for context-grounded responses.
- Developed a Streamlit chat interface with doctor and patient modes, conversation history, source references, and emergency symptom detection for responsible and transparent medical assistance.

FAST Sports Complex Management System (FSCM)

Python, LangGraph, LangChain, FastAPI, Next.js, Groq (LLaMA 3), Tavily API, ChromaDB

- Built a full-stack platform using React, Node.js, and PostgreSQL to manage facilities, events, and equipment with secure CRUD operations and role-based dashboards.
- Designed and deployed RESTful APIs and a responsive frontend, enabling real-time data updates and seamless user interaction across the system.

EDUCATION

BS Computer Science *2023 – 2027*

FAST – National University of Computer and Emerging Sciences, Karachi, GPA: 3.02

HSSC Pre-Engineering *2021 – 2023*

IBA Community College, Khairpur – Aga Khan Examination Board